

Quadrotor Cascade PID Controller Automatic Tuning

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Abstract—Integrating systems are commonly encountered in many applications, among them are the unmanned aerial vehicles. This paper proposes an automatic tuning scheme for cascade PID control of integrating systems with application to unmanned aerial vehicles. Using relay feedback control in a closed-loop setting, the auto-tuner systematically finds the system's closed-loop frequency response, open-loop frequency response and integrator plus delay model. PID controller parameters are subsequently determined using a set of simple tuning rules. Because the tuning rules contain the desired closed-loop time constant as a performance specification together with desired gain and phase margins, the auto-tuner designed is naturally suited for a cascade PID control system where the inner-loop control system typically has much faster response speed than the primary control system, which can be readily achieved by adjusting the two desired time constants. Experimental results obtained from the auto-tuner application on a quadrotor Unmanned Aerial Vehicles (UAV) are used to demonstrate the efficacy of the proposed approach.

I. INTRODUCTION

Automatic tuning of PID control has been the pursuit of a generation of control engineers for the past several decades ([1], [2], [3] [11], [24], [14]). With the functionality of automatic tuning, a PID controller will have the capability of setting its parameters with minimal human intervention.

It is well understood that a key component among all auto-tuners is the application of system identification tools to find the mathematical representation of the physical systems. There are many papers published in the control engineering literatures for automatic tuning of PID controllers. The following discussions only give a glimpse of the vast literatures. [11] suggested the use of a set of pseudo-random-binary-sequence (PRBS) signal together with cross-correlation technique to identify the critical information needed for auto-tuner design. [23] proposed to use the Fast Fourier Transform (FFT) to obtain the plant frequency response, the step response and a second order model with time delay for automatic tuning of PID controllers. [18] proposed an auto-tuning algorithm based on relay feedback control experiments by minimizing the load disturbance effects and maximizing the integral gain with minimum constraints on the gain and phase margins. [25] extended PID auto-tuning method to applications in the area of fraction order plus time delay system. [10] proposed the use of nonlinear optimization techniques to find the PID controller parameters in a multi-loop controller tuner. [13] extended the gain and phase margin design for PID control of single-input and single-output system to multi-input and multi-output system. [7] proposed an auto-tuner for nonlinear and multivariable systems.

This paper proposes a relay feedback auto-tuning algorithm for cascade integrator with delay systems. Because integrating systems are on the boundary of instability, it requires stabilization before a relay experiment can be safely conducted. It appears that the majority of the auto-tuners are derived for stable systems only ([12], [14]) and the use of stabilizing controllers on integrating systems is seldom considered. This paper brings together several existing approaches in the areas of system identification and PID controller design to form a new auto-tuner for integrating systems. The auto-tuner has four key components that differentiate it from the other auto-tuners. The first is that the relay experiment is conducted for an integrating system under closed-loop control, for the reason that the open-loop system is unstable; the second is that frequency sampling filters are used to recursively estimate the fundamental frequency of the closed-loop system ([22], [21]); the third is the simple computation of the frequency response of the integrating system and the associated dynamic model; and the fourth is the PID controller design using the tuning rules for integrator with delay systems presented in ([20], [21]). Because the PID controller design method requires the user to specify a desired closed-loop time constant as the performance parameter, the auto-tuner designed is then naturally suited for tuning a cascade PI control system where the inner-loop control system typically has a much faster dynamic response speed than the outer-loop response, which can be readily achieved by adjusting the two desired time constants.

In this paper, the key motivation for designing the auto-tuner comes from the application of PID control of quadrotor UAVs. Control system design and implementation is paramount in the applications of quadrotor UAVs [16] for it is an unstable system because of its physical structure. It is known that the linearized model of a quadrotor UAV contains a double integrator together with actuator dynamics. The most popular control strategy in practice ([6], [5] and [17]) is a cascade PID control system, which provides the basis for the application of the auto-tuner designed in this paper. Automatic tuning of PID controllers for a quadrotor UAV not only saves time, but also ensures safe operation, preventing it from crashing in the process of finding the controller parameters.

The remainder of this paper is outlined as follows. In Section II, the auto-tuner is designed with detailed steps on its implementation. In Section III, the quadrotor dynamics is discussed and a cascade PI control system is configured. Experimental results are presented in Section IV followed by the conclusions of this paper.

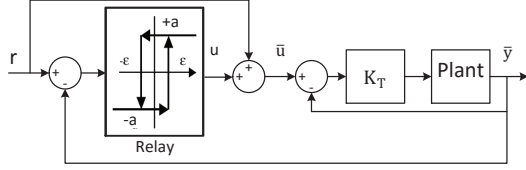


Fig. 1. Block diagram of relay feedback control.

II. DESIGN OF AUTO-TUNER FOR PID CONTROLLERS

This section presents the design of auto-tuner for PID control of integrating systems, which consists of four components: relay feedback control, recursive estimation of the frequency response, estimation of the integrating plus delay model and PID controller design. When the auto-tuner is applied to tune a cascade control system, the algorithm remains the same for both inner-loop and outer-loop controllers.

A. Relay Feedback Control

In the auto-tuner design, a proportional controller with known gain K_T is used to stabilize the integrating system and a relay feedback control system is deployed for the output of the closed-loop system. Figure 2 shows the configuration of the closed-loop system under relay feedback control. The reference signal $r(t)$ is a constant that represents the steady-state operation of the plant. ϵ is the hysteresis selected to avoid the possible random switches caused by the measurement noise and a is the amplitude of the relay. The signal $\bar{y}(t)$ represents the actual output measurement. The closed-loop control is at the steady-state operation before the relay feedback control is switched on. To ensure a bumpless transfer to the relay feedback control, the same reference signal $r(t)$ is used as the reference signal for the relay-feedback control. Assuming that at the sampling time t_0 the relay feedback control is enabled. The sampling interval is defined as $\Delta t = t_k - t_{k-1}$. To initialize the relay element, at the sampling time t_{-1} , $\bar{u}(t_{-1}) = r(t_{-1})$. For all $t_k \geq t_0$, the calculation of the reference signal to the closed-loop control system uses the following relay switching rules:

- 1) Calculate the relay feedback error: $e(t_k) = r(t_k) - \bar{y}(t_k)$.
- 2) If $|e(t_k)| \leq \epsilon$; then $\bar{u}(t_k) = \bar{u}(t_{k-1})$.
- 3) If $|e(t_k)| > \epsilon$; then $\bar{u}(t_k) = r(t_k) + a \times \text{sign}(e(t_k))$.

This relay feedback control system will produce a sustained oscillation illustrated by the experimental data in Figure 5 ([4], [1]).

Assume that the period of the oscillation is T . It is well known that the frequency of the periodic signal $\bar{u}(t)$, denoting by $\omega_1 = \frac{2\pi}{T}$, approximately has $-\frac{\pi\epsilon}{4a}$ as the imaginary part.

B. Recursive Estimation of Frequency Response

To estimate the closed-loop frequency response

$$T(j\omega_1) = \frac{K_T G(j\omega_1)}{1 + K_T G(j\omega_1)}$$

where $G(j\omega_1)$ is the open-loop frequency response at ω_1 , the pair of input and output signals corresponding to the relay feedback control system is used. That is, the input signal equals the relay output signal:

$$u(t) = \bar{u}(t) - r(t) = a \times \text{sign}(e(t))$$

and the closed-loop output signal with steady-state removed becomes

$$y(t) = \bar{y}(t) - r(t) = -e(t)$$

For a period T , the Fourier series expansion of the periodic input signal $u(t)$, is expressed as [15]

$$u(t) = \frac{4a}{\pi} \left(\sin \frac{2\pi}{T} t + \frac{1}{3} \sin \frac{6\pi}{T} t + \frac{1}{5} \sin \frac{10\pi}{T} t + \dots \right) \quad (1)$$

By choosing sampling interval Δt and the number of samples within one period $N = \frac{T}{\Delta t}$, the discretized input signal $u(t)$ at sampling instant $t_k = k\Delta t$ becomes

$$u(k) = \frac{4a}{\pi} \left(\sin \frac{2\pi k}{N} + \frac{1}{3} \sin \frac{6\pi k}{N} + \frac{1}{5} \sin \frac{10\pi k}{N} + \dots \right) \quad (2)$$

Equation (2) has an alternative expression as

$$u(k) = \frac{2a}{j\pi} \left(e^{j\omega_d k} + \frac{1}{3} e^{j3\omega_d k} + \frac{1}{5} e^{j5\omega_d k} + \dots \right) - \left(e^{-j\omega_d k} + \frac{1}{3} e^{-j3\omega_d k} + \frac{1}{5} e^{-j5\omega_d k} + \dots \right) \quad (3)$$

where for notational simplicity, $\omega_d = \frac{2\pi}{N}$ denoting the fundamental frequency in discrete-time.

For a stable system with transfer function $T(z)$, in general, it has the z-transfer function model in frequency sampling filter form:

$$T(z) = \sum_{l=-\frac{N-1}{2}}^{\frac{N-1}{2}} T(e^{jl\omega_d}) F^l(z), \quad (4)$$

where $F^l(z)$ is the l th frequency sampling filter given by

$$F^l(z) = \frac{1}{N} \frac{1 - z^{-N}}{1 - e^{jl\omega_d} z^{-1}} = \frac{1}{N} (1 + e^{jl\omega_d} z^{-1} + \dots + e^{j(N-1)l\omega_d} z^{-(N-1)}).$$

The output of the closed-loop control system under relay feedback with input $u(k)$ is expressed as

$$y(k) = \sum_{l=-\frac{N-1}{2}}^{\frac{N-1}{2}} T(e^{jl\omega_d}) f^l(k) + v(k) \quad (5)$$

where $f^l(k)$ is the output of the l th frequency sampling filter and $v(k)$ is the output measurement noise assumed to be Gaussian distributed with zero mean and variance σ^2 .

If the sampling filters are assumed to have zero initial conditions then the outputs of the frequency sampling filters in response to the input signal $u(k)$ satisfy the following relationships:

$$f^l(k) = \begin{cases} 0, & \text{if } l = 0, \pm 2, \pm 4, \pm 6, \pm 8, \dots \\ \frac{2a}{j\pi|l|} e^{jl\omega_d k}, & \text{if } l = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9, \dots \end{cases} \quad (6)$$

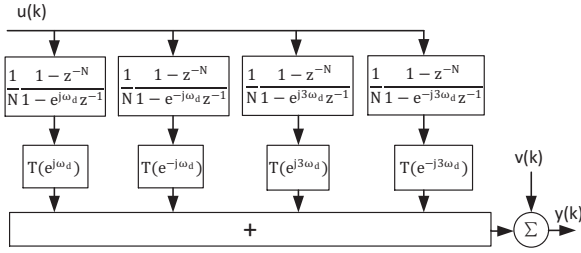


Fig. 2. Block diagram of frequency sampling filter model using relay control.

Equation (5) is approximated using the first two pairs of frequency sampling filters while neglecting the rest of the terms. That is

$$y(k) \approx T(e^{j\omega_d})f^1(k) + T(e^{-j\omega_d})f^{-1}(k) + T(e^{j3\omega_d})f^3(k) + T(e^{-j3\omega_d})f^{-3}(k) + v(k) \quad (7)$$

Define the complex parameter vector to be estimated as

$$\theta = [T(e^{j\omega_d}) \ G(e^{-j\omega_d}) \ T(e^{j3\omega_d}) \ T(e^{-j3\omega_d})]^T$$

and its corresponding regressor vector as

$$\phi(k) = [f^1(k) \ f^{-1}(k) \ f^3(k) \ f^{-3}(k)]^T$$

where A^{T*} denotes the complex conjugate transpose of A .

The recursive least squares estimation algorithm is used to estimate the frequency response parameters, $T(e^{j\omega_d})$, $T(e^{-j\omega_d})$, $T(e^{j3\omega_d})$, $T(e^{-j3\omega_d})$, under relay feedback control. Here, a standard recursive least squares algorithm is written as

$$P(k-1) = P(k-2) - \frac{P(k-2)^T \phi(k) \phi(k)^T P(k-2)}{1 + \phi(k)^T P(k-2) \phi(k)} \quad (8)$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + P(k-1) \phi(k) (y(k) - \phi(k)^T \hat{\theta}(k-1)) \quad (9)$$

where $P(-1)$ and $\hat{\theta}(0)$ are the initial conditions selected for the recursive least squares algorithm. $\hat{\theta}(k)$ contains the estimated frequency response parameters.

C. Estimation of Integrating Plus Delay Model

With the knowledge of the proportional controller K_T , the frequency response of the plant $G(e^{j\omega_d})$ is calculated from the closed-loop frequency response relationship,

$$T(e^{j\omega_d}) = \frac{G(e^{j\omega_d})K_T}{1 + G(e^{j\omega_d})K_T}$$

leading to

$$G(e^{j\omega_d}) = \frac{1}{K_T} \frac{T(e^{j\omega_d})}{1 - T(e^{j\omega_d})} \quad (10)$$

Note that the discrete-time frequency response $G(e^{j\omega_d})$ is a close approximation to its continuous-time counterpart under the assumption that the system operates in a fast sampling environment, where the equivalent continuous-time frequency

is $\omega_1 = \frac{\omega_d}{\Delta t}$. Letting $G_p(j\omega_1)$ denote the plant continuous-time frequency response at the fundamental frequency ω_1 , it follows that

$$G_p(j\omega_1) \approx G(e^{j\omega_d})$$

For an integrating plus time delay system, a single frequency is sufficient to determine its gain K_p and time delay d . The approximate model of an integrating system is assumed to be of the following form:

$$G_p(s) = \frac{K_p e^{-ds}}{s} \quad (11)$$

For most physical systems, there are more or less approximations involved in obtaining the integrator plus time delay model.

Now, letting the frequency response of the integrator plus delay model (11) be equal to the estimated $G_p(j\omega_1)$ leads to

$$\frac{K_p e^{-jd\omega_1}}{j\omega_1} = G_p(j\omega_1) \quad (12)$$

Equating the magnitudes on both side of (12) gives

$$K_p = \omega_1 |G_p(j\omega_1)| \quad (13)$$

where $|e^{-jd\omega_1}| = 1$. Additionally, from (12), the following relationship holds:

$$e^{-jd\omega_1} = \frac{j\omega_1 G_p(j\omega_1)}{K_p}$$

This gives the estimate of time delay as

$$d = -\frac{1}{\omega_1} \tan^{-1} \frac{\text{Imag}(jG_p(j\omega_1))}{\text{Real}(jG_p(j\omega_1))} \quad (14)$$

It is seen here that if the system is approximated by an integrator with time delay, the plant information at a single frequency is sufficient to determine the plant gain and time delay.

Once the integrator plus time delay model is obtained, the PID controller parameters are calculated using the tuning rules presented in [20] and [21]. Although there are many design methods available for integrator plus time delay model for instance, [19], the set of tuning rules adopted here has the characteristics of being simple and robust, which was developed using frequency response analysis without approximation of time delay. With the tuning rules, the PID controller parameters are calculated using the following normalized parameters:

$$K_c = \frac{\hat{K}_c}{dK_p}$$

$$\tau_I = d\hat{\tau}_I$$

$$\tau_D = d\hat{\tau}_D$$

For damping coefficient $\xi = 1$, the normalized PID controller parameters are calculated as

$$\hat{K}_c = \frac{1}{0.5080\beta + 0.6208} \quad (15)$$

$$\hat{\tau}_I = 1.9885\beta + 1.2235 \quad (16)$$

$$\hat{\tau}_D = \frac{1}{1.0043\beta + 1.8194} \quad (17)$$

where the parameter β is the scaling factor for the desired closed-loop time constant, which is defined as

$$\tau_{cl} = \beta d$$

III. APPLICATION TO QUADROTOR CONTROL

The quadrotor dynamics is described by Euler's equation of motion ([6], [8], [9]):

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} (I_{yy} - I_{zz})qr/I_{xx} \\ (I_{zz} - I_{xx})pr/I_{yy} \\ (I_{xx} - I_{yy})pq/I_{zz} \end{bmatrix} + \begin{bmatrix} 1/I_{xx} & 0 & 0 \\ 0 & 1/I_{yy} & 0 \\ 0 & 0 & 1/I_{zz} \end{bmatrix} \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} \quad (18)$$

where I_{xx} , I_{yy} and I_{zz} are the moments of inertia for the three axes in x , y , z directions, p , q and r their angular velocities and τ_x , τ_y , τ_z are the corresponding torques. Furthermore, the relationship between Euler angular velocities and the body frame angular velocities (p , q and r) are described in the following differential equation:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi)/\cos(\theta) & \cos(\phi)/\cos(\theta) \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \quad (19)$$

In order to fly a UAV, the closed-loop control of the three Euler angles ϕ , θ and ψ is necessary. Therefore, there are three outputs in the control system. The manipulated variables or the control signals are the three torques, τ_x , τ_y , τ_z , along the x , y and z directions. The body frame angular velocities p , q and r along the x , y and z directions are the intermittent variables in the cascade control system which is to be configured in the sequel.

The quadrotor is assumed to have symmetric structure with four arms aligned with the x -axis and y -axis, and as a result there is no interaction between the torques along the three axes. Because of the products of angular velocities in (18) and the sinusoidal functions in (19), the quadrotor system is a nonlinear system.

When the quadrotor is working in a normal operating condition with a balanced load, the angular velocities p , q and r have a steady-state operating condition at 0, which has a linearized expression as, for example,

$$qr \approx q_0 r_0 + r_0(q - q_0) + q_0(r - r_0) = 0$$

where the steady-state values of $q_0 = r_0 = 0$. With this assumption, (18) is simplified to

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} 1/I_{xx} & 0 & 0 \\ 0 & 1/I_{yy} & 0 \\ 0 & 0 & 1/I_{zz} \end{bmatrix} \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} \quad (20)$$

Additionally, the steady-state operating conditions for the three Euler angles are chosen to be zero, leading to the linearized dynamics model for (19) as

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (21)$$

In summary, the dynamics models for a quadrotor UAV consist of two sets of integrators in a normal operating condition given by (20)-(21).

Fig 3 shows a cascade feedback control structure of one axis. The inner-loop controller is to control the secondary plant, where its reference signal is the desired angular velocity that is also the control signal generated from the outer-loop (primary) controller. For the cascade control system, the primary objective is to control the primary plant to achieve desired closed-loop performance.

IV. EXPERIMENTAL VALIDATION

The quadrotor consists of five main components: RC transmitter/receiver, IMU sensor board, data logger, micro processor and actuators. The RC transmitter/receiver is to send and receive reference signals. The IMU sensor board is to measure the Euler angles and angular velocities. The data logger is to record flight data such as Euler angles and reference signals. The micro-processor is to generate control signals to stabilize the UAV's attitude. Actuators are to generate thrust and torques, which consist of motor drives, DC motors, gearboxes and blades.

In order to conduct identification experiments in a controlled environment, the quadrotor is required to be fixed on a mechanical stance for the testing on ground. For the identification purpose, the test-bed shown in Fig. 4 is built and used to conduct the relay experiments for identification of the quadrotor's 3-axis dynamics.

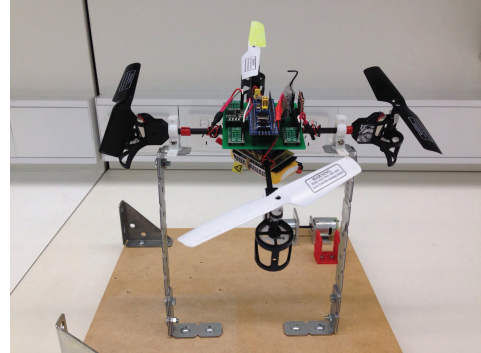


Fig. 4. Quadrotor test-bed

A. Auto-tuning of Secondary PI Controller

Before the relay control experiment, a proportional controller $K_{T1} = 0.06$ is selected to stabilize the secondary system with sampling interval $\Delta t = 0.01$ (sec). The sensor measurement of velocity, $\dot{\theta}(t)$, contains noise. Thus, a relay amplitude of 0.8 together with a hysteresis level of 0.1 is selected to reflect the measurement noise level. Figure 5 shows a segment of the relay feedback control data. From the input signal to the inner-loop closed-loop control system, the period of the sustained oscillations is identified as $N = 28$ samples, leading to the fundamental frequency in the frequency sampling filter as $\frac{2\pi}{N}$. The use of the frequency sampling filter based estimation algorithm gives the inner-loop closed-loop frequency response as

$$T(e^{j\frac{2\pi}{N}}) = -1.225 - j0.5137$$

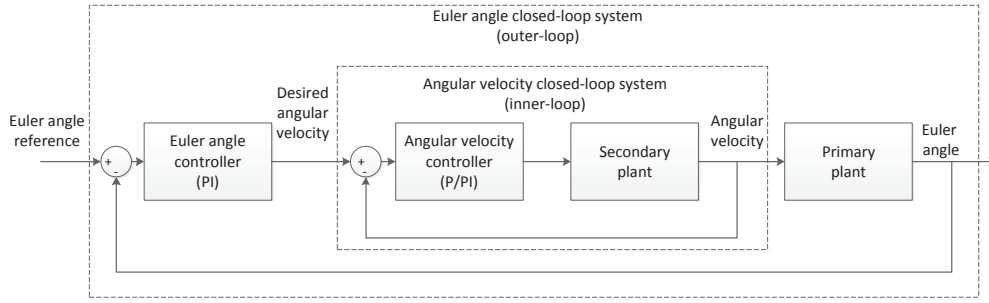


Fig. 3. Cascade feedback control structure

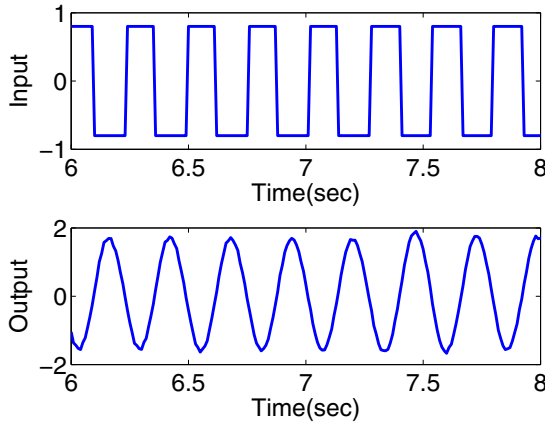


Fig. 5. Relay feedback control signals from inner-loop system: top figure input signal; bottom figure output signal.

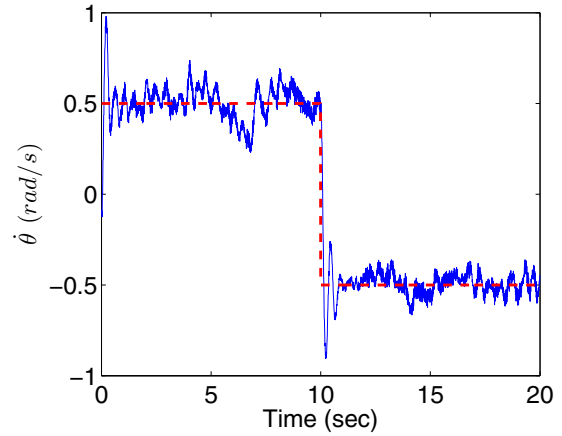


Fig. 6. Inner-loop step response in closed-loop control. Dashed line: reference signal; solid line: output.

Converting this discrete-time frequency to continuous time frequency, which is $\omega_1 = \frac{2\pi}{N\Delta t} = 22.44$, together with the knowledge of the proportional controller used in the relay experiment ($K_{T_1} = 0.06$), the continuous-time frequency response of the inner-loop plant is calculated as

$$G_1(j\omega_1) = \frac{1}{K_{T_1}} \frac{T(e^{j\frac{2\pi}{N}})}{1 - T(e^{j\frac{2\pi}{N}})} = -9.551 - j1.6421$$

From this frequency information, an integrating plus delay model is identified as

$$G_1(s) = \frac{217.5589e^{-0.0624s}}{s}$$

By choosing the desired closed-loop time constant as $\tau_{cl} = \beta d = 3d = 0.1872$ (sec) and damping coefficient of 1, from the tuning rules shown in (15)-(17), the PI controller parameters are found for the inner-loop control system as $K_c = 0.0343$ and $\tau_I = 0.4487$. This set of PI controller parameters approximately gives a gain margin of 3 and phase margin of 48 degree for the closed-loop system with the integrator plus delay model. Figure 6 shows the closed-loop step response of $\dot{\theta}(t)$ where the reference signal has a magnitude of 0.5 (rad/sec). It is seen from this figure that the closed-loop velocity response follows the reference signal

without steady-state error, and there is a large overshoot and a slight oscillation. Additionally, there are disturbances and measurement noise in the inner-loop system. For a cascade control system, the inner-loop control system is required to have a fast response speed, which is achieved in the design here.

B. Auto-tuning of Primary PI Controller

The proportional controller used for the relay control experiment is selected as $K_{T_2} = 2$. The amplitude of the relay is 0.4 and the hysteresis level ϵ is 0.05 to prevent the relay from random switching. Figure 7 shows a segment of the input and output data generated from this relay feedback control. The averaged period of the sustained oscillation is $N = 59$ in number of samples, which gives the fundamental frequency in discrete-time as $\frac{2\pi}{59}$. Frequency sampling filters model is used to estimate the closed-loop frequency response based on the set of input and output data shown in Figure 7, yielding to

$$T(e^{j\frac{2\pi}{N}}) = -0.1317 - j0.3130$$

With the proportional controller $K_{T_2} = 2$, the frequency response of the outer-loop system is found at $\omega_1 = \frac{2\pi}{N\Delta t} =$

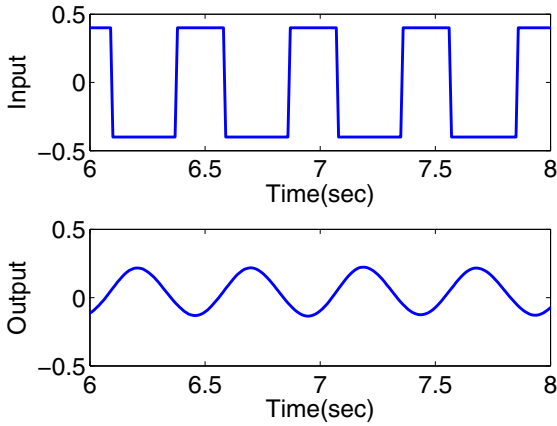


Fig. 7. Relay feedback control signals from outer-loop system: top figure input signal; bottom figure output signal.

10.649 as

$$G_2(j\omega_1) = \frac{1}{K_{T_2}} \frac{T(e^{j\frac{2\pi}{N}})}{1 - T(e^{j\frac{2\pi}{N}})} = -0.0896 - j0.1135$$

From this frequency information, the integrator plus delay model for the primary system is calculated as

$$G_2(s) = \frac{1.54e^{-0.0627s}}{s}$$

For a typical cascade control system design, the outer-loop control system should have a slower desired closed-loop response than that of the inner-loop control system. By selecting the desired closed-loop time constant as $\tau_{cl} = \beta d = 8.5d$ with damping coefficient $\xi = 1$ ($\beta = 8.5$, gain margin ≈ 7 , and phase margin ≈ 63 degree), where d is the delay of the integrating system, the PI controller parameters are computed using the tuning rules

$$K_c = 2.095; \quad \tau_I = 1.137$$

For comparison purpose, when a faster desired closed-loop time constant $\tau_{cl} = 5d$ ($\beta = 5$, gain margin ≈ 5 , and phase margin ≈ 55 degree) and a slower desired closed-loop time constant $\tau_{cl} = 10d$ ($\beta = 10$, gain margin ≈ 9 , and phase margin ≈ 65 degree) are used in the computation, the PI controller parameters become $K_c = 3.27$, $\tau_I = 0.7006$, and $K_c = 1.8155$, $\tau_I = 1.3244$.

V. CONCLUSIONS

This paper proposed the design of an auto-tuner for cascade control of integrating systems using the relay feedback control mechanism. The key steps in the auto-tuner include identification of the critical frequency information of the closed-loop feedback control system using frequency sampling filter model, the calculation of the open-loop frequency response, obtaining an approximate integrator plus delay model and the application of PID controller tuning rules. A direct application of the auto-tuning algorithm is shown in the cascade control of a quadrotor UAV, followed by experimental evaluation on a laboratory test-bed.

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